OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 25 OF 55

Multilingual Speech Emotion Recognition Using Deep Learning

Domain: Audio Signal Processing & Deep Learning | **Module:** project_25_speech_emotion

OFFICIAL ABSTRACT

Human speech communicates not only words but also emotions through variations in tone, pitch, intensity, rhythm, and speaking style. Conventional speech recognition systems mainly focus on converting speech into text and often overlook the emotional state of the speaker. The proposed Multilingual Speech Emotion Recognition System Using Deep Learning aims to identify emotions from spoken audio across multiple languages, enabling more intelligent and human-aware voice interaction. The system accepts speech as input and extracts important acoustic features such as pitch, frequency, energy, and spectral characteristics. Deep learning techniques are then used to learn the relationship between these speech patterns and emotional states such as happiness, sadness, anger, fear, and neutrality. The model is designed to work across multiple languages, making it suitable for multilingual environments where pronunciation, vocabulary, and speaking styles may differ. The major uniqueness of the proposed system is its focus on how a person speaks rather than only what the person says. The system can identify emotional characteristics even when the language changes. It can also be extended to estimate emotion intensity, allowing the system to distinguish between low, medium, and high levels of an emotion instead of providing only a basic emotion label. This makes the solution more meaningful for real-world applications. Another important feature is the possibility of cross-language emotion learning, where emotional patterns learned from one language can assist recognition in another language. Audio preprocessing techniques can also be incorporated to improve recognition when speech contains background noise. With further development, the system can support real-time emotion detection through microphone input. The proposed system can be applied in intelligent virtual assistants, customer-support systems, online education, entertainment, and human-computer interaction. By combining multilingual speech processing, language-independent emotional feature learning, and emotion-intensity recognition, the project aims to create a more context-aware voice intelligence system. Overall, the project

> demonstrates how deep learning can bridge the gap between speech recognition and emotional understanding, enabling AI systems to recognize not only what a person says, but also the emotional context in which it is spoken.

📌 Project Summary & System Specifications

Project Title	Multilingual Speech Emotion Recognition Using Deep Learning
Specialization / Category	Audio Signal Processing & Deep Learning
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5025`
Local Implementation	`project_25_speech_emotion`